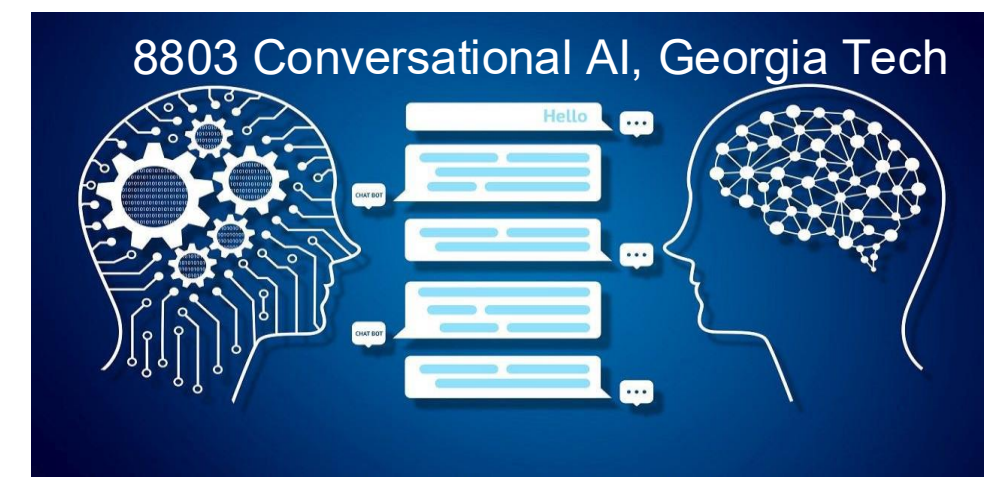


“Where Does the Model Look?” — Attention Across Modalities in Vision-Language Conversation

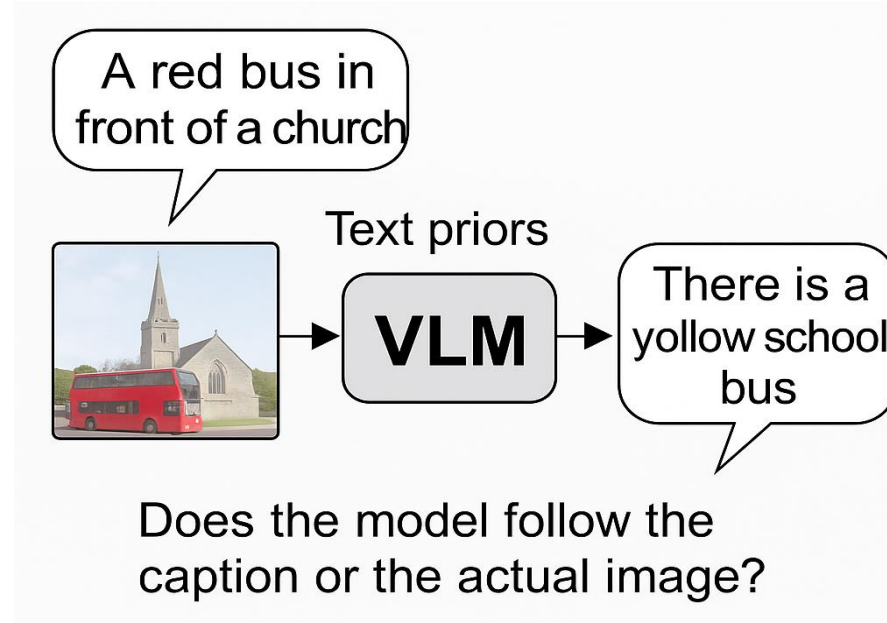
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Analyzing visual grounding and conversational bias in BLIP-2, LLaVA-1.5, and Qwen2.5-VL

Problem

- VLMs can discuss images but may rely on text priors instead of really “looking” at the pixels.
- Real use is multi-turn dialogue, so wrong captions or early answers can bias later turns.
- Most benchmarks use single image–question pairs missing,
 - how misleading captions shift answers
 - how performance changes as questions become more visually demanding.
- BLIP-2, LLaVA-1.5, and Qwen2.5-VL use vision and language differently but are rarely compared in one controlled conversational setup.



Key Questions:

- > When does a VLM actually use the image instead of predominantly following textual priors?
- > How do caption accuracy, question difficulty, and dialogue history affect hallucination and visual grounding?

Background

- Task:** analyze conversational grounding in vision–language models (VLMs).
- Setup:** COCO image I , caption C , and ladder questions $L0$ – $L4$.
- Visual Necessity Ladder:**
 - $L0$ is language-only
 - $L1$ – $L4$ are increasingly image-dependent.
- Prior Work:** mostly single image–question or captioning tasks; attention often only viewed qualitatively
- Research Gap:** little controlled comparison of architectures or conversational context on visual grounding.

Methods

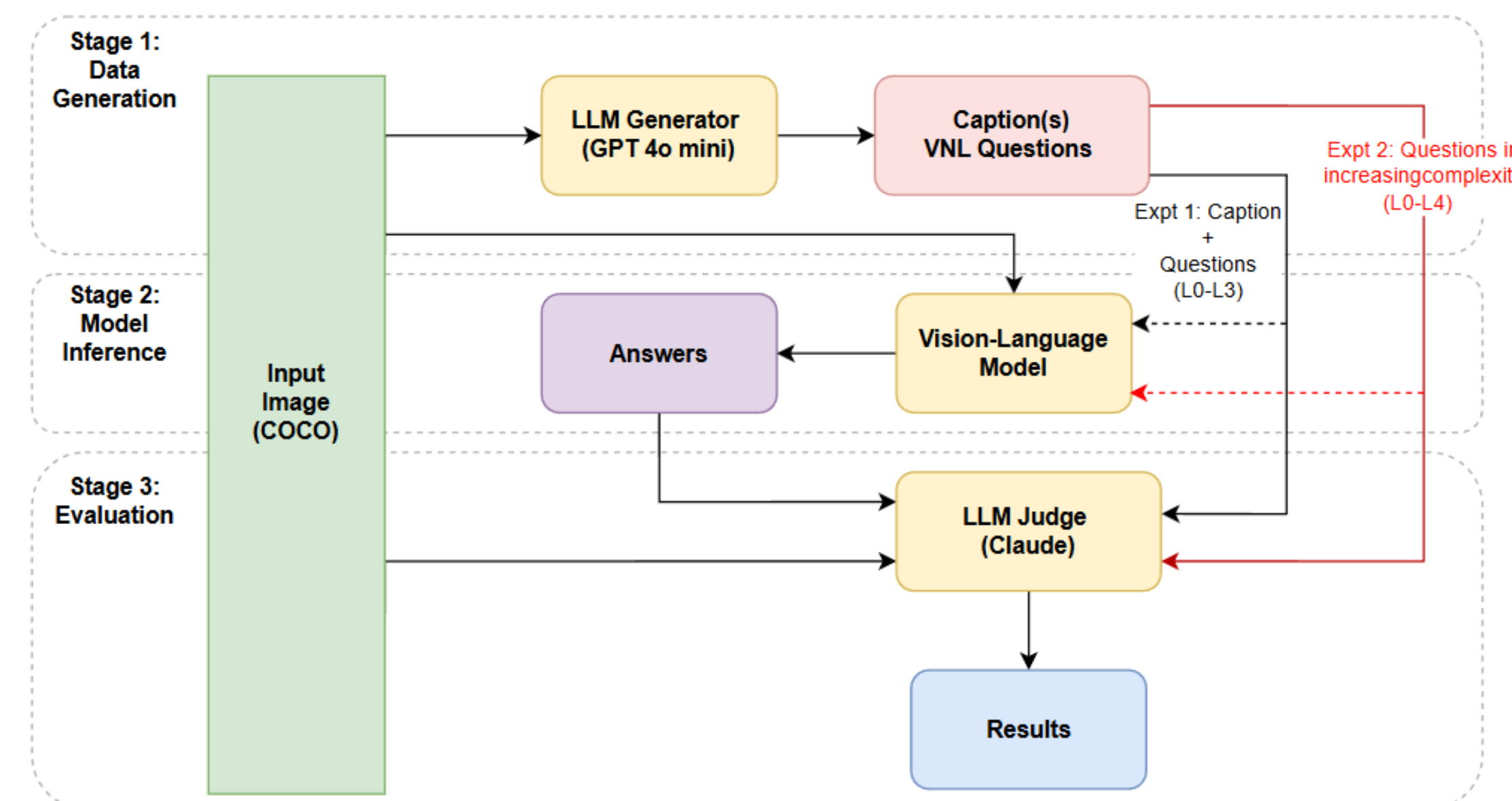


Figure 1. Experimental Evaluation Frame

Experiment 1: Caption Sensitivity



Questions:

- L0:** What is a common outdoor recreational activity?
- L1:** What type of animal is catching the toy?
- L2:** What color is the toy being caught?
- L3:** Why is it important for pets to exercise outdoors?

Correct Caption: A brown and white dog catches a red frisbee outdoors

Incorrect Caption: A black and white cat catches a blue ball outdoors.

Correct Caption Answers:

- L0:** playing frisbee
- L1:** a dog
- L2:** red
- L3:** it is good for their health

Incorrect Caption Answers:

- L0:** catching a ball
- L1:** dog
- L2:** blue
- L3:** A dog can run faster than a cat

Experiment 2: Incremental Reasoning

Caption: A brown and white dog jumps catching a red and green frisbee

Questions:

- L0:** What animal is commonly kept as a pet?
- L1:** What color is the dog in the image?
- L2:** What colors are on the frisbee the dog caught?
- L3:** Is the dog touching the ground or in the air?
- L4:** What action is the dog performing with the frisbee?

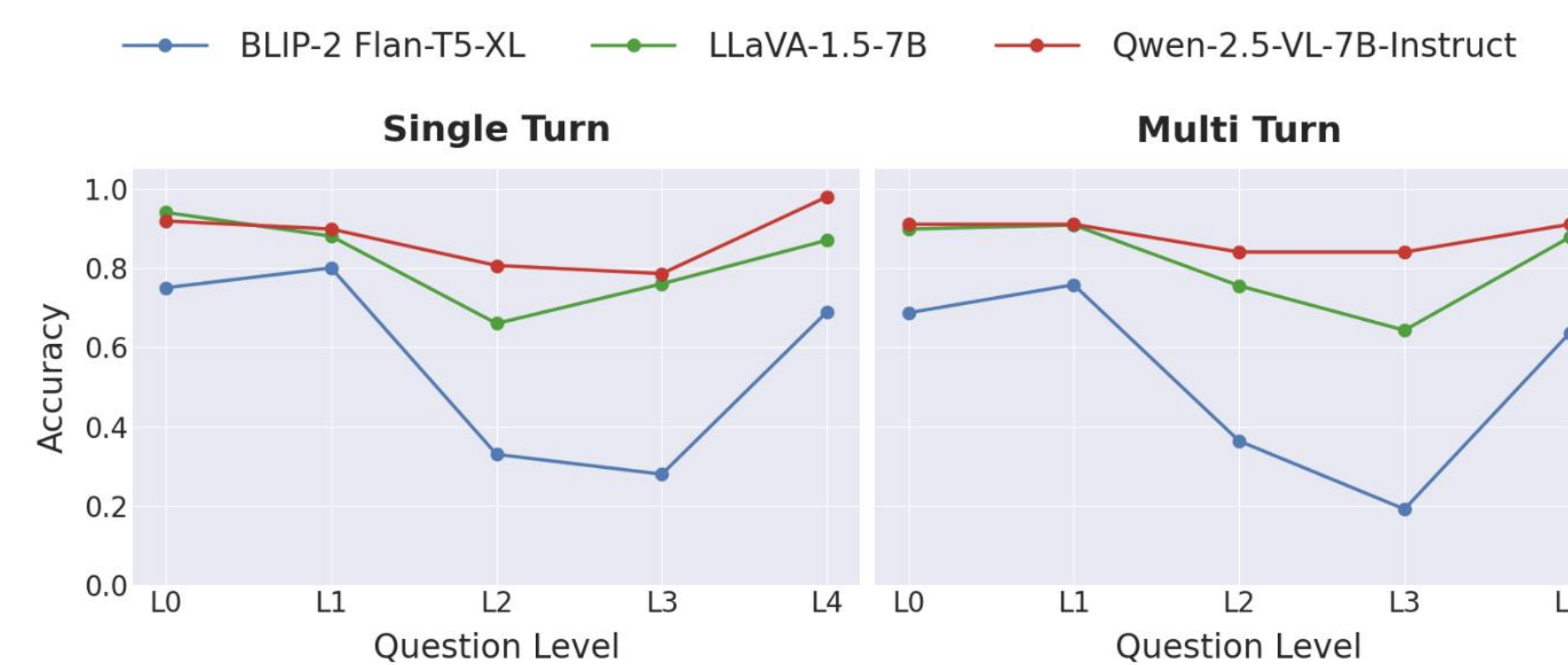
Answers:

- L0:** dog
- L1:** brown and white
- L2:** red and green
- L3:** Yes
- L4:** catching

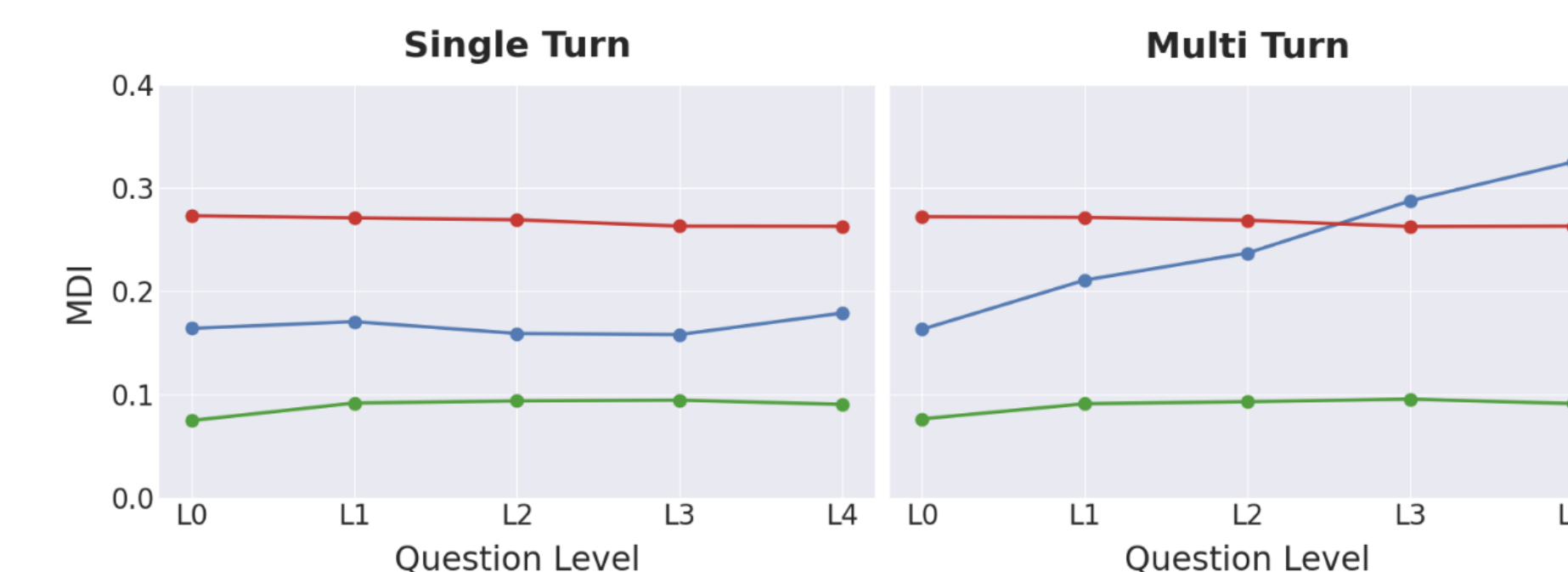
Results and Analysis

Experiment 2 Results:

Model Accuracy Across Question Difficulty and Turn Types



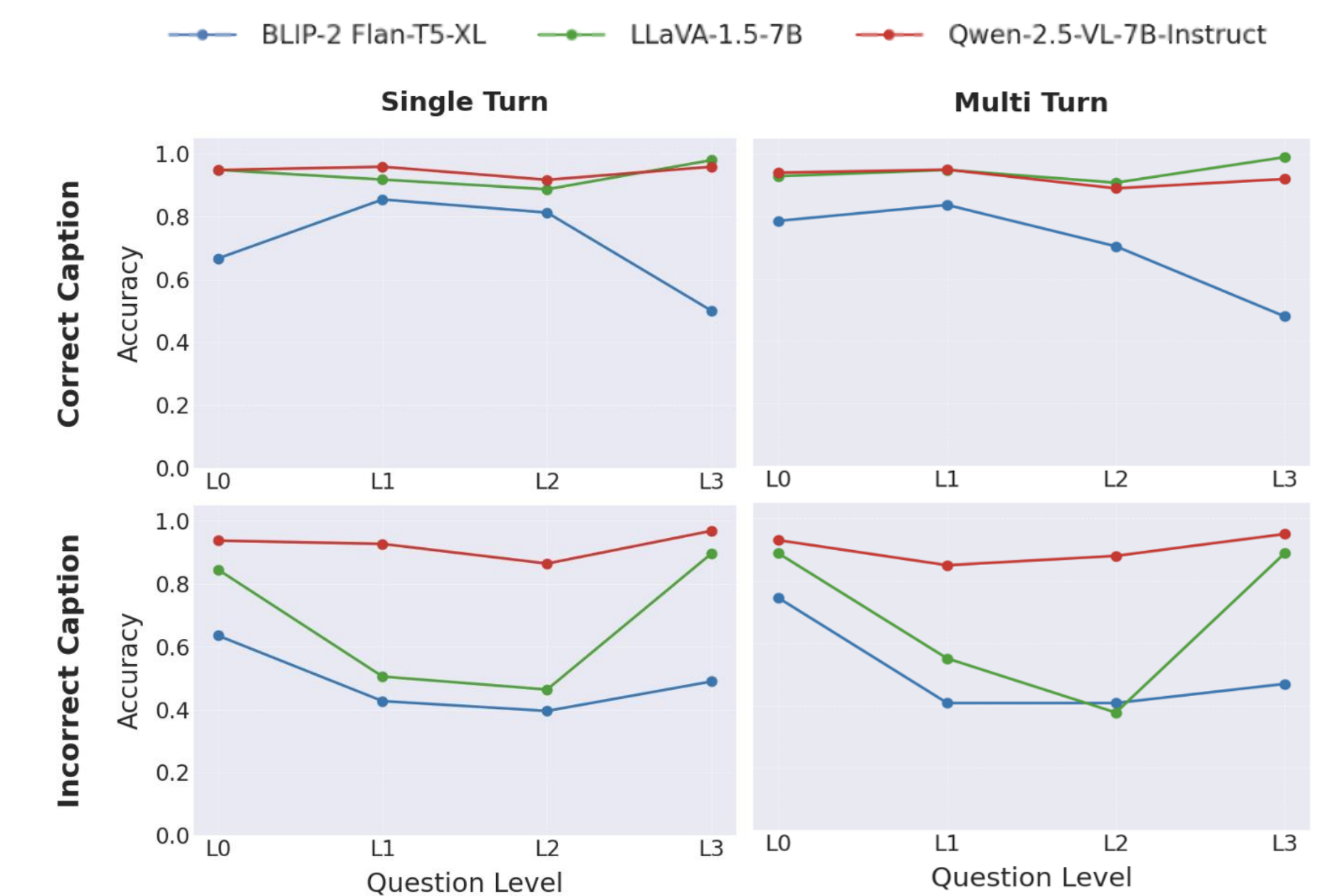
Modality Dominance Index (MDI) Across Question Difficulty and Turn Types



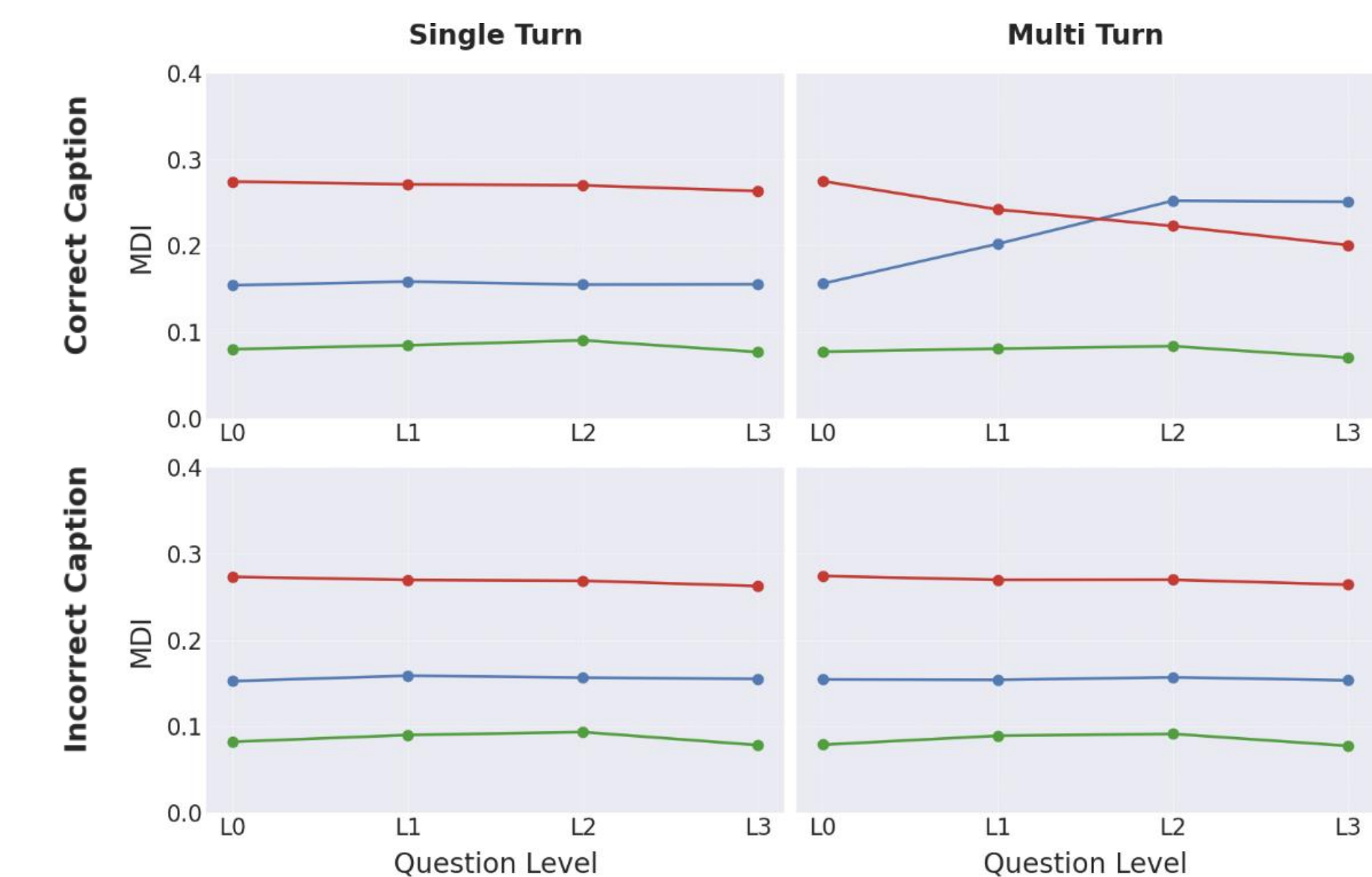
Results and Analysis

Experiment 1 Results:

Model Accuracy Across Caption Types and Turn Types



Modality Dominance Index (MDI) Across Caption Types and Turn Types



Conclusion

- Accuracy is highest on generic L0 questions and drops on mid-ladder, image-heavy questions.
- Incorrect captions hurt performance much more than correct ones, especially in multi-turn dialogue.
- As questions incrementally become more visually-based, attention shifts slightly towards images but hallucinations remain.
- Conversational history can amplify early caption errors, so grounding over multiple turns is still fragile.